

Demonstrating Z-Band: Enabling Subtle Hand Interactions with Bio-impedance Sensing on the Wrist

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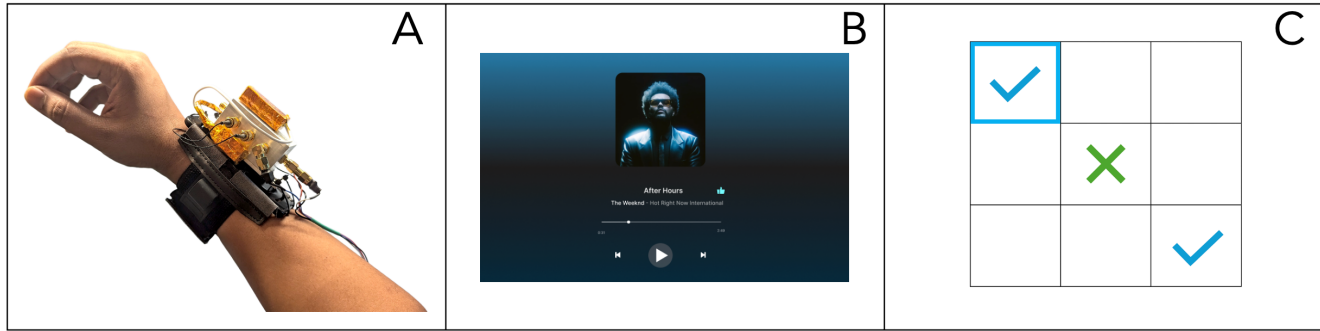


Figure 1: (A) The Z-Band prototype worn on the wrist. (B) A screenshot of the music player app. (C) A screenshot of the tic-tac-toe game. Both apps are controlled by Z-Band gesture-based interaction.

ABSTRACT

The increasing popularity of wearable extended reality (XR) technologies presents unique challenges for user input, as traditional methods like touchscreens or controllers can be cumbersome and less practical. As a result, researchers are exploring novel approaches to enable seamless and intuitive interaction within XR environments. In this work, we present Z-Band, a novel interaction device that enables subtle finger input in a wrist-worn form factor. Z-Band utilizes radio frequency (RF) sensing, leveraging the human hand as an antenna to detect subtle changes in hand impedance caused by finger movements. We demonstrate the feasibility and effectiveness of Z-Band by showcasing its capabilities through two applications: a music player controlled by finger gestures and a gesture-based game.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**; **Ubiquitous and mobile devices**; **Interaction techniques**.

KEYWORDS

Wearable input devices, AR/VR, RF sensing, Sensor fusion, Machine Learning, Interaction Techniques

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1 INTRODUCTION

The advent of extended reality (XR) technologies, encompassing virtual reality (VR), augmented reality (AR), and mixed reality (MR), has ushered in a new era of immersive digital experiences. As these technologies become increasingly integrated into our daily lives through wearable devices, the importance of seamless and intuitive interaction mechanisms has grown exponentially. Subtle input, encompassing a range of discreet interaction techniques that minimize disruption to the user's natural behavior and environment, has emerged as a critical research area for enhancing the user experience in XR scenarios. The significance of subtle input in XR interactions is amplified when considering the context of wearable devices. These devices are designed for continuous use, often in social and public settings, where overt gestures or interactions may be undesirable or impractical.

Previous approaches for subtle hand input in XR have relied on external sensors like cameras, requiring users to keep their hands within the sensor's field of view, thus limiting mobility and

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natural interaction. Researchers have also explored embedding sensors directly into wearable devices, utilizing technologies such as cameras, ultrasound, and inertial measurement units (IMUs). However, these methods often face challenges in user adaptation and real-time performance, mainly when aiming to capture subtle gestures [1–3].

Z-Band leverages radio frequency (RF) sensing to track finger movements by injecting RF signals into the hand, which acts as an antenna. Z-Band measures changes in hand impedance caused by subtle finger movements. Using a wide frequency sweep, the system captures even the most minute gestures. While this approach has been demonstrated in ring wearables [4], this work extends it to wrist-worn devices, addressing the unique challenges associated with sensing at a greater distance from the interaction site.

To overcome signal attenuation, Z-Band introduces a novel RF architecture that enables amplification of the transmitted signal while maintaining a return path for the reflected signal. The challenge of reduced SNR is tackled by utilizing multiple sensing channels and incorporating additional sensors like IMUs. This multi-faceted approach allows Z-Band to accurately capture finger movements in a wrist-worn form, opening up new possibilities for subtle and intuitive interactions in wearable technology.

2 IMPLEMENTATION

The Z-Band prototype (Figure 1 A) uses hand impedance data obtained through S11 reflection coefficient measurements input from a portable Vector Network Analyzer (VNA). To overcome signal attenuation issues, Z-Band incorporates a unique RF architecture. This architecture includes two circulators and an amplifier, enabling amplification of the transmitted signal and accurate measurement of the reflected signal from the body.

RF circulators are passive, three-port devices that direct signal flow in a specific, non-reciprocal manner. The signal from the VNA enters the first circulator, which then directs it to an *amplifier*, which sends the amplified signal to the second circulator and couples it to the body. The reflected signal from the body returns through the second circulator, passes through an attenuator to protect the VNA, returns to the first circulator, and is finally directed back to the VNA for measurement and analysis. This configuration enables Z-Band to achieve precise hand impedance measurements with increased signal power.

The VNA sweeps a frequency range of 70 to 210 MHz, experimentally determined to be optimal for propagation through the body. An RF switch directs the transmitted signal to three distinct channels on the wrist, enabling the capture of bio-impedance data at different locations to gain spatial information. The wrist prototype also incorporates two IMU units for additional sensing capabilities. The setup is connected to a Raspberry Pi, which facilitates data transmission to a laptop for further processing and analysis.

3 INTERACTIONS AND APPLICATIONS

Z-Band facilitates subtle yet expressive thumb-to-index finger gestures, capitalizing on the thumb's dexterity and the two fingers' close proximity. The system supports six distinct gestures: swipes in four cardinal directions (up, down, left, right), a single tap, and a

double tap. We developed two applications to showcase Z-Band's utility: a music player and a tic-tac-toe game (Figure 1 B and C).

The music player app lets users navigate through their music library using directional swipes and play or pause songs with taps. The tic-tac-toe game lets players select their desired spot on the board through swipes and place their mark (X or tick) with taps. These demonstrations highlight the potential of Z-Band for intuitive and unobtrusive interaction in various contexts.

4 CONCLUSION

Z-Band introduces a novel approach to enabling subtle finger input on wrist-worn devices using RF sensing with a unique sensing architecture. By leveraging the hand as an antenna and analyzing changes in hand impedance, Z-Band successfully captures subtle finger movements, addressing the limitations of traditional input methods in wearable XR contexts. The implementation and demonstration of gesture-controlled applications highlight the feasibility and potential of Z-Band for intuitive, unobtrusive interaction in various scenarios.

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